When Does Training-Free DTW Match Learned Models on Hand Landmarks? A Data-Efficiency and Signer-Independence Study of Isolated Sign Recognition

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Abstract

We ask when matching a sign's hand landmarks against stored examples with dynamic time warping (DTW) is competitive with trained neural networks for isolated sign recognition. On PopSign ASL v1.0 (American Sign Language, smartphone video) with the dataset's signer-disjoint split, we compare DTW and Euclidean nearest neighbour, DTW prototypes, a 1D-CNN, a BiGRU and a small Transformer on the same MediaPipe features, varying the examples per sign from 1 to all and the vocabulary from 10 to 50 signs. The analysis was pre-registered, each method family chose its own preprocessing on validation signers, and intervals come from resampling the 8 test signers. Contrary to our pre-registered expectation, the learned models beat DTW at every number of examples per sign we tried: at 50 signs and 10 examples per sign the CNN scores 61.1% top-1 against 38.6% for DTW, and with all training data 85.9% against 63.4%. Richer features and preprocessing raised DTW to at best 71.3% with all data. We report latency, training cost, ablations and limitations. This is an isolated-sign study on one dataset; it is not translation and makes no claim about benefit to Deaf users.

1 Introduction

Isolated sign recognition assigns a label from a fixed vocabulary to a short video of one sign. It is a small part of sign language processing and is not translation: it ignores grammar, context and most non-manual signals. We study it because it is a clean setting in which to ask a practical methods question. When there are only a few labelled examples per sign, does matching a new video against stored examples with dynamic time warping (DTW) work as well as a trained neural network, and at what point does the network pull ahead?

A common recipe extracts hand and body landmarks with MediaPipe (Lugaresi et al., 2019; Zhang et al., 2020), turns them into per-frame features, and either trains a sequence model or compares sequences directly. Nearest-neighbour DTW needs no training and can be updated by adding examples, which is attractive for small, changing vocabularies. In general time-series classification, nearest-neighbour DTW with a tuned warping window is a strong baseline (Bagnall and Lines, 2014), and deep networks are competitive on large benchmarks (Ismail Fawaz et al., 2019). We have not found a controlled comparison of the two on signer-independent sign data that varies both the number of examples per sign and the vocabulary size, so we ran one.

What this paper does. On PopSign ASL v1.0 (Starner et al., 2023), with the dataset's signer-disjoint split, we compare DTW nearest-neighbour matching, DTW prototypes, Euclidean nearest neighbour, a 1D-CNN, a BiGRU and a small Transformer, all on the same MediaPipe features and the same queries. We vary the references per sign from 1 to all, the vocabulary from 10 to 50 signs, the feature set, the handling of missing hands and the DTW variant. The analysis was written down before the test data were scored (`PREREG.md`), and uncertainty is estimated by resampling signers.

What it does not do. It does not propose a new method, claim state-of-the-art accuracy, or show that the systems help Deaf people. It covers one sign language (American Sign Language), one dataset, smartphone selfie video, and a mostly one-handed vocabulary. The work was done by hearing researchers without a Deaf co-author; Section 7 discusses what that limits.

Relation to earlier work by the same author. An earlier conference paper (Mankar et al., 2024) described a MediaPipe and DTW sign recogniser built on an open-source implementation

(gabguerin, 2022). That paper compared DTW with an LSTM on a small custom dataset without signer-independent evaluation. The present study keeps that pipeline's feature idea as one baseline and replaces its evaluation.

2 Related work

Datasets. Isolated-sign datasets include WLASL (Li et al., 2020) (over 2000 words, over 100 signers) and MS-ASL (Joze and Koller, 2018) (1000 signs, over 200 signers); ASL Citizen (Desai et al., 2023) (2731 signs, 52 signers), collected from consenting contributors; PopSign (Starner et al., 2023), recorded on smartphones; LSA64 (Ronchetti et al., 2023) for Argentinian Sign Language; and LSFB-ISOL (Fink et al., 2021) for Belgian French Sign Language. We use PopSign because its licence permits derived data, its signers consented to a public database, and it ships a signer-disjoint split.

Landmark-based recognition. Skeleton and pose-based models include graph convolutional networks (Jiang et al., 2021) and a pose-based Transformer (Bohacek and Hruz, 2022). These are trained models evaluated on their own benchmarks; our smaller models are not meant to reproduce them, only to represent the learned side of a data-efficiency comparison.

DTW and template matching for signs. DTW (Sakoe and Chiba, 1978) has several variants: banded and approximate forms (Salvador and Chan, 2007), averaged prototypes (DBA, Petitjean et al., 2011) and the differentiable soft-DTW (Cuturi and Blondel, 2017). For signs, Arcanjo et al. (2024) apply FastDTW to angular features computed from MediaPipe landmarks, and Saggio et al. (2020) compare k-NN with DTW against a CNN on sensor-glove data. Vandendriessche et al. (2025) study one-shot recognition against a sign dictionary with learned embeddings. We did not find prior work that varies references per sign and vocabulary size together on signer-independent video data; our search was not exhaustive. The features we call F1 follow the hand-angle vectors of the open-source implementation (gabguerin, 2022) used in our earlier work (Mankar et al., 2024).

Ethics of sign language technology. Interdisciplinary and Deaf-led work argues that sign language AI has often been shaped by hearing researchers' priorities, rests on unrepresentative datasets and overstates communication benefits (Bragg et al., 2019, 2021; Desai et al., 2024). A survey of Deaf community members reports their views on automatic translation (Tran et al., 2023). We take these as constraints on how we scope and word this paper.

3 Data and methods

Data. We use the `game` category of PopSign ASL v1.0 (Starner et al., 2023): 250 signs recorded on smartphones, CC BY 4.0, with a training split of 31 signers, a validation split of 8 and a test split of 8. The code checks that no signer appears in two splits. Vocabularies of 10, 50, 100 and 250 signs are nested prefixes of a fixed random order of the signs. A vocabulary is reported only if every one of its signs was processed; others are marked as not run. Videos are downloaded, converted to landmarks and deleted.

Landmarks and features. We use MediaPipe Holistic 0.10.14 (Lugaresi et al., 2019) on frames resized to a 480 pixel short side. Coordinates are scaled by frame width and height before any angle is computed (the earlier implementation used normalised coordinates, which distorts angles on portrait frames). A sequence is trimmed to the span between the first and last frame with a detected hand. Feature set F1 is the 210 pairwise angles between the 21 hand connection vectors per hand (identical information to the earlier 21×21 angle matrix; Euclidean distances differ by a constant factor). F2 is wrist-centred, scale-normalised hand shape. F3 combines F1 and F2, F4 adds hand location relative to the shoulders, F5 adds upper-body pose and F6 adds face-contour landmarks. The main grid uses F1. Missing hands are zero-filled in the main setting.

Methods. **DTW 1-NN** compares a query with every stored reference using DTW with Euclidean local cost and a Sakoe–Chiba band chosen on validation signers, and returns the label of the nearest reference. **Euclidean 1-NN** compares sequences resampled to 32 frames. **DBA prototype** replaces each sign's references by their DTW barycentre (length equal to the median reference length). The learned models are a 1D-CNN, a two-layer Bi-GRU and a three-layer Transformer encoder (each about 0.5M parameters) trained with cross-entropy for a fixed number of gradient steps with no early stopping, so no validation labels enter train-

ing. DTW distances are computed by our own compiled implementation, which is tested against tslearn (Tavenard et al., 2020). Every method family chooses its own preprocessing on validation signers only (50 signs, 10 examples per sign, one draw), then it is fixed: how missing hands are handled (zero-fill, carry the last detection forward, or drop those frames), whether features are divided by their standard deviation over the references, and the DTW band (3, 5, 8 frames or none) for DTW; learning rate (10^{-3} or 3×10^{-4}) and epoch budget (20 or 80) for the learned models. Training takes $\min(\max(\lceil \text{epochs} \cdot N/64 \rceil, 200), 4000)$ gradient steps for N training examples, so the largest conditions get only a few epochs.

Protocol. References (or training examples) per sign $k \in \{1, 2, 5, 10, 20, \text{all}\}$ are drawn at random from the training split, nested across k, with three draw seeds (which also seed network initialisation). Test queries are up to three videos per test signer and sign, drawn once and shared by all methods. A query on which no hand is ever detected is counted as an error. We report top-1 accuracy with 95% intervals from resampling the 8 test signers (10,000 resamples, percentile intervals; cf. Field and Welsh, 2007), and paired differences use the same resamples. With only 8 clusters the intervals are wide and we do not report p-values. Two hypotheses were fixed in advance and judged at 97.5% (§4); all other comparisons are exploratory.

4 Results

All numbers in this section are generated from the stored result files (`CLAIMS.md` lists each one). Intervals are 95% unless stated, from resampling the 8 test signers. Seeds are averaged first. "Examples per sign" k means references for the nearest-neighbour methods and training examples for the learned models; "all" uses every training video of the sign.

4.1 Main comparison

At a vocabulary of 50 signs and 10 examples per sign, the 1D-CNN reaches 61.1% top-1 (54.1–69.6), the Transformer 57.0% and the BiGRU 56.7%, against 38.6% (30.7–48.4) for DTW 1-NN and 42.7% for Euclidean 1-NN (Table 1, Figure 1). With all training videos the learned models reach 85.9% (Transformer), 85.5% (CNN) and 84.9% (BiGRU), and DTW 63.4%. At 10 signs the ordering is the same: 93.2% (CNN) and 85.1% (DTW) with all training videos.

Method	1	2	5	10	20	all
DTW 1-NN	17.2	22.1	31.0	38.6	45.9	63.4
Euclidean 1-NN	18.9	24.7	35.0	42.7	47.5	63.2
DBA prototype	17.2	15.2	13.4	13.1	12.7	12.4
1D-CNN	23.2	32.0	48.2	61.1	71.6	85.5
BiGRU	20.0	28.7	43.9	56.7	67.9	84.9
Transformer	23.7	30.3	45.9	57.0	66.2	85.9
BiGRU + aug.	21.9	28.5	44.5	57.9	67.7	85.1

Table 1: Top-1 accuracy (%) at a vocabulary of 50 signs on the 8 held-out test signers, by examples per sign (feature set F1, preprocessing chosen per method on validation signers). Intervals are in `results/tables/main_top1.md`.

4.2 Pre-registered hypotheses

H1 is not supported. We expected DTW to beat each learned model at vocabulary 50 with $k \leq 5$. Instead every learned model is above DTW at $k =$ 1, 2 and 5: 9 of the nine paired differences have 97.5% intervals wholly on the learned model's side of zero (`results/tables/hypotheses.csv`). For example, at $k = 1$ the CNN is 6.0 points above DTW (95% interval 4.2 to 7.7), and at $k = 5$ it is 17.2 points above (13.9 to 21.0). **H2 is supported**: with all training videos each learned model is above DTW by about 22.2 (CNN), 22.5 (Transformer) and 21.5 (BiGRU) points, and the intervals remain above zero when adjusted for three comparisons (yes).

4.3 No crossover in the tested range

We looked for the number of examples per sign at which learned models overtake DTW and found none: at both vocabularies the best learned model is at or above DTW from $k = 1$ upwards (smallest such k: 1 at 10 signs, 1 at 50). In 36 of the 36 comparisons of a learned model with DTW (3 models, 6 values of k, 2 vocabularies) the 95% interval excludes zero, and in 36 the learned model's point estimate is higher (exploratory; no multiplicity correction). The gap is smallest at $k = 1$, where every method is weak: at 50 signs no method exceeds 23.7% (Transformer, the best) with one example per sign. Euclidean 1-NN is not worse than DTW at 50 signs: it is ahead by 2.7, 4.0 and 4.1 points at $k = 2$, 5 and 10 (intervals exclude zero) and indistinguishable at $k = 1$, 20 and "all" (1.8, 1.6 and -0.2 points). At 10 signs DTW is clearly ahead of Euclidean 1-NN (interval excluding zero) only

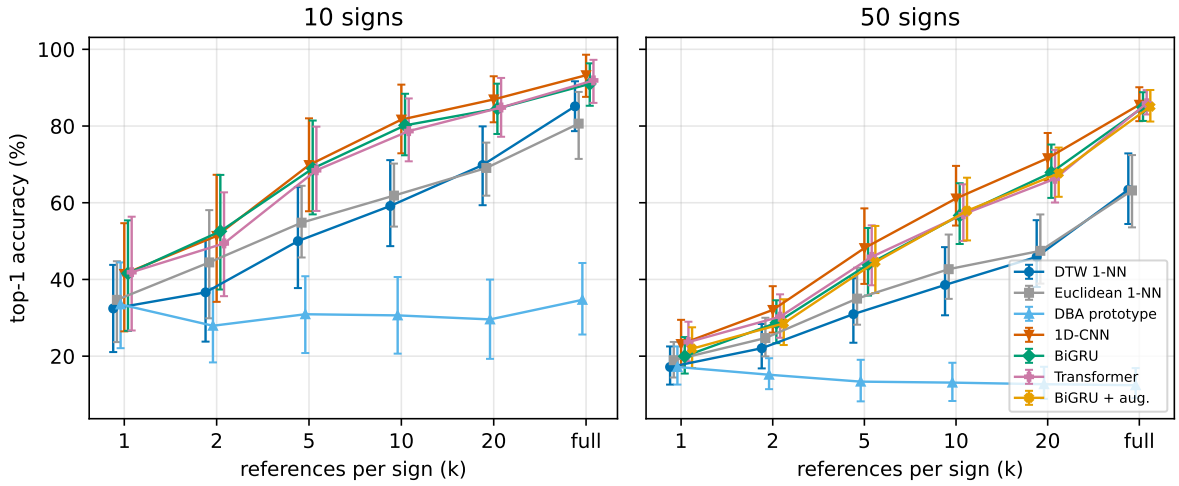


Figure 1: Top-1 accuracy versus examples per sign for vocabularies of 10 and 50 signs. Bars are 95% intervals from resampling test signers.

with all training videos (Euclidean minus DTW: -4.5 points, interval -9.0 to -1.3), while Euclidean 1-NN is ahead at $k = 2$ (7.8 points).

4.4 DTW variants

DBA prototypes (one barycentre per sign) are poor and get worse with more references per sign at 50 signs (17.2% at $k = 1$, 12.4% with all videos), so we do not recommend averaging signer variants into a single template. On one fixed draw (50 signs, 10 examples per sign, seed 0), exact DTW scores 37.2%, FastDTW (radius 1) 38.4% and the original repository's pipeline 39.1%; the intervals overlap completely (28.6–51.8% for the original pipeline), so these variants are not distinguishable at this sample size. FastDTW is not faster here: our compiled exact DTW takes 10.8 ms per query against 500 references on one thread, while the pure-Python FastDTW takes 1198.1 ms, so the difference reflects the implementations, not the algorithms.

4.5 Preprocessing matters more for the template-matching baselines

On validation signers (50 signs, 10 examples per sign), DTW with the naive default of zero-filled missing hands and raw features scored 29.0%, against 40.4% for the configuration chosen on validation (frames without a hand dropped, standardised features, band of 8 frames). The learned models were much less sensitive: best validation accuracy with zero-filled gaps versus the chosen setting was 64.4% vs 66.6% for the CNN. Every family chose to drop frames without a detected hand. We give each family its own validation-selected preprocessing so that no method is held back by an arbitrary default; the naive default would have made the DTW baseline look 11.4 points worse on validation.

Setting (features/missing/hand slots)	$k = 5$ DTW	$k = 5$ CNN	$k =$ all DTW	$k =$ all CNN
F1/zero/slots	23.9	45.6	57.0	85.5
F1/ffill/slots	26.9	43.7	59.5	82.7
F1/drop/slots	31.0	48.2	63.4	85.5
F1/zero/dominant	28.0	59.8	54.3	88.6
F2/zero/slots	18.0	40.4	48.5	89.7
F3/zero/slots	27.2	53.2	66.7	90.3
F4/zero/slots	27.4	53.1	67.2	93.3
F5/zero/slots	27.6	52.3	67.1	92.9
F6/zero/slots	12.4	35.4	39.8	92.6
F3/drop/slots†	36.1	55.0	70.6	89.8
F4/drop/slots†	36.3	55.4	71.3	92.6
F5/drop/slots†	34.7	52.5	70.9	93.3
F4/drop/dominant†	38.1	66.0	67.1	93.4

Table 2: Top-1 accuracy (%) at 50 signs by feature set, missing-hand handling and hand slots, for DTW 1-NN and the CNN. Rows without a dagger were pre-registered and run with the stated settings; rows marked † are an exploratory supplement added after seeing the main results (see `PREREG.md`). The F1/drop row is the main setting. Other methods and intervals: `results/tables/ablation.md`.

4.6 Feature sets, missing hands and handedness (ablations)

Table 2 shows the effect of each factor (the pre-registered feature ablations used zero-filled missing hands, so compare rows within the table).

Dropping frames without a hand helps DTW (23.9% with zero-fill to 31.0% at $k = 5$) more than the CNN (45.6% to 48.2%). Putting the more active hand in the first slot (dominant-hand canonicalisation) raises the CNN at $k = 5$ from 45.6% to 59.8% and DTW from 23.9% to 28.0%, but with all training videos DTW goes down (57.0% to 54.3%). Adding hand location (F4) raises DTW with all videos from 57.0% (F1) to 67.2% and the CNN from 85.5% to 93.3%; adding upper-body pose changes little; adding face contours lowers DTW sharply (39.8% with all videos) while the CNN stays near 92.6%.

Across every setting we ran, the best DTW result was 38.1% at $k = 5$ and 71.3% with all videos, against 66.0% and 93.4% for the best CNN setting, so no feature or preprocessing choice we tried closes the gap. These maxima are over many settings on the test signers and are therefore optimistic for both families.

4.7 Augmentation, one- and two-handed signs, and per-sign errors

Light augmentation (time stretch, noise, scale) changed the learned models by at most a few points: for the CNN at 50 signs, 25.0% versus 23.2% at $k = 1$ and 85.8% versus 85.5% with all videos. The learned models' lead over DTW is therefore not an artefact of leaving them unaugmented. We could not compare one- and two-handed signs: under the pre-registered rule (both hands detected in at least half of the hand-present frames), 0 of the 50 signs are two-handed and 50 are one-handed, consistent with PopSign's one-handed focus.

With all training videos, DTW is below 50% on 14 of the 50 signs, with the lowest being sleepy (0.0%), whereas the CNN is below 50% on 0 signs (lowest: police, 50.0%). The most frequent DTW confusion is dryer recognised as dry (10 test queries). PopSign has no phonological annotation, so we make no claim about why signs are confused; the full lists are in `results/tables`.

4.8 Cost: training and inference

Template matching needs no training but its inference cost grows with the number of references: with 500 references (10 per sign) one query takes 10.8 ms (95th percentile 34.9 ms) on one thread of an Apple M4 Max with our compiled DTW, and 547.6 ms with all 21876 training videos as references (Euclidean nearest neighbour: 2.0 and 91.0 ms). A forward pass of a trained model takes 0.13 ms (CNN), 0.25 ms (Transformer) and 0.93 ms (BiGRU) on one CPU thread, independent of the number of training examples. Training the CNN and Transformer took on average 3 and 4 seconds at 10 examples per sign and 24 and 26 seconds with all videos on the Apple GPU (BiGRU: 109 and 283 seconds; training times varied with load on the shared machine). Latency was measured with other processes running, and no low-power device was available. These costs exclude MediaPipe landmark extraction (8.8 wall-clock hours for the 50 signs on this machine).

4.9 Coverage

Vocabularies of 100 and 250 signs were not run: landmark extraction for 50 signs took 8.8 hours on this machine and the larger vocabularies would have taken several times longer. The ASL Citizen replication and soft-DTW prototypes in the preregistration were not run. Of the 1038 test queries at 50 signs, 1 had no detected hand and is counted as an error; the training and validation splits have 17 and 3 such videos.

5 Discussion

What the data say. On PopSign with signer-independent splits and hand-angle features, small learned models beat nearest-neighbour DTW from one example per sign upward, at vocabularies of 10 and 50 signs, and the two pre-registered expectations came out as: H1 (DTW ahead with few examples) not supported, H2 (learned models ahead with all data) supported. With one example per sign and 50 signs every method is weak (the best point estimate is 23.7%). Euclidean nearest neighbour was never worse than DTW at 50 signs, and DBA prototypes were poor. Preprocessing mattered a great deal for the template-matching baselines (11.4 points on validation between the naive and the chosen setting) and little for the learned models, which is a reason to tune baselines before concluding anything about them.

Why DTW trails is untested. Several explanations are plausible and we did not test them: the per-frame angle features are noisy, so extra alignment freedom may bring unrelated signs closer; the learned models can weight informative dimensions while DTW weights all dimensions equally; and signer variation, which a model trained on 31 signers can absorb, is a direct cost for template

matching. Adding hand location and the preprocessing choices narrowed the gap only partly. We therefore make no causal claim about the mechanism.

Relation to other results. In general time-series classification, tuned nearest-neighbour DTW is a strong baseline (Bagnall and Lines, 2014). Our result does not contradict that finding, because the setting differs: 420-dimensional noisy features from landmark detection, 50 classes, and test signers unseen in training; we did not investigate how far each of these differences matters. The earlier conference paper from the same author (Mankar et al., 2024) reported DTW ahead of an LSTM on a small custom dataset without a signer-independent split; we cannot re-run that experiment, and the present signer-independent results do not support carrying its conclusion over to this setting.

Practical reading, with limits. For this dataset and these features, a small learned model is the better default once a few labelled examples per sign and a few seconds of training are available, and it is also much cheaper at inference when many reference videos exist. Template matching keeps its practical advantages of requiring no training and of letting a new sign be added by storing one example, but in our experiments that convenience cost accuracy. This is a statement about one dataset, one language and our feature choices (Section 6); it says nothing about whether any of these systems is useful to Deaf people, and accuracy on isolated signs is not communication access (Section 7).

6 Limitations

- **One dataset, one language.** All results are for ASL on PopSign (47 signers, selfie phone video, mostly one-handed signs). We do not test other datasets, languages, cameras or continuous signing.
- **Few signers.** The test split has 8 signers, so signer-clustered intervals are wide and small differences are not resolvable. Training noise (3 seeds) is smaller than signer variation.
- **Small models, limited tuning.** The learned models are small and tuned on a small grid at one condition (vocabulary 50, 10 examples per sign). All three chose the longest epoch budget in the grid, so longer training might help them further, and the step cap of 4000 gives the largest conditions only a few epochs. Larger models or pre-training may change the picture. We include one augmentation arm to test sensitivity at one vocabulary only.
- **Only F1 in the main grid, and a handicap risk for DTW.** The main comparison uses the hand-angle features of the earlier implementation, which carry no hand location or orientation. DTW and Euclidean matching each chose among the preprocessing options listed in Section 3 on validation signers, but other options, local costs or learned metrics might favour template matching more.
- **Landmark quality.** Results depend on MediaPipe 0.10.14 and on the resize used; the fraction of videos or frames without detected hands is reported in the tables. Face and other non-manual signals are mostly not modelled.
- **Not translation, not an application.** Top-1 accuracy on isolated signs is not communication access. We did not evaluate any system with Deaf users.
- **No phonological analysis.** PopSign has no phonological annotation; we report data-driven confusions and make no claim about phonological similarity.
- **Timing.** Latency was measured on one Apple M4 Max laptop with other processes running; no low-power device was available, so embedded performance is not measured.
- **Pre-registration scope.** The plan was written by the author before test scoring, but is not a registered report and was not independently time-stamped; the git history is the record.

7 Ethics and broader impact

Who did the work. This study was designed and run by hearing researchers without a Deaf co-author. Deaf-led analysis of sign language AI argues that such work often centres hearing researchers' priorities and a "communication barrier" framing (Desai et al., 2024). We chose a methods comparison on an existing, consented dataset rather than an application, and we make no claim that any result benefits Deaf people.

Data. PopSign participants consented to inclusion in a public database, and the dataset card lists identification and appearance-based uses as out of scope; we keep face landmarks local, release only hand and pose landmarks, and never attempt to identify signers. The dataset card also discourages continuous-signing and translation claims, which we avoid.

Risks. A cheap recogniser could be presented as a substitute for interpreters or language access. High accuracy on 250 isolated signs does not support that, and we discourage it. Community perspectives on automatic sign language translation (Tran et al., 2023) and dataset governance (Bragg et al., 2021) should be consulted before any deployment.

Community feedback. [To be completed by the author before submission: state whether Deaf community members reviewed the framing of this paper, what changed as a result, or that no such review took place.]

8 Reproducibility

Code, configuration, pinned requirements, per-query results and generated tables are in the accompanying repository (`make quick` reruns everything; `make report` regenerates every table, figure and number in this paper from the stored results). Landmark extraction versions and timings are logged, and every result file has a provenance file with the commit and library versions. The preregistration is `PREREG.md`; deviations from it are listed there. Derived hand and pose landmarks are released under CC BY 4.0 with attribution to Georgia Tech and the Deaf Professional Arts Network; videos and face landmarks are not released.

Use of AI tools. An AI coding assistant (Claude, Anthropic) helped write code, tests and a first draft of the text and checked references. The author specified the design, reviewed and ran the code, checked every number against the generated results files and every citation against its publisher or arXiv record, and takes responsibility for the content.

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